

1. Introduction

This project designs a recommendation system to suggest SHL assessments based on natural-language hiring requirements. Recruiters often describe roles in free text, making keyword matching insufficient. The solution uses semantic understanding of both job descriptions and assessment metadata.

2. Data Collection

Source: SHL Product Catalog (Individual Test Solutions only).

Custom crawler fetched 377 assessments with name, URL, and type; stored in CSV.

3. Representation & Embeddings

Metadata: assessment name, test type (Knowledge, Personality), keywords.

Model: SentenceTransformer (all-MiniLM-L6-v2) for semantic embeddings.

4. Retrieval & Indexing

Embeddings indexed using FAISS for efficient similarity search.

Query enriched, converted to embedding, top-3K retrieved, then re-ranked.

5. Recommendation Logic

Balanced mix of:

Knowledge/Technical tests

Personality/Behavioral tests

Ensures relevance and diversity for real-world hiring workflows.

6. Evaluation

Dataset: labelled_train.xlsx

Metric: Recall@10 (URL match).

Average recall computed across queries.

7. Results

Modest Recall@10 due to small dataset, strict URL matching, and general-purpose embeddings.
System shows correct semantic retrieval and practical recommendations.

8. Deployment

Local web app using FastAPI + Jinja2.

Features: input hiring requirement, view recommendations, direct links.

9. Limitations & Future Work

Domain-specific fine-tuning

Hybrid keyword + semantic ranking

Learning-to-rank

Advanced skill extraction

10. Conclusion

A complete, reproducible solution for recommending SHL assessments using semantic search.

Meets assignment requirements and reflects real-world recruitment workflows.